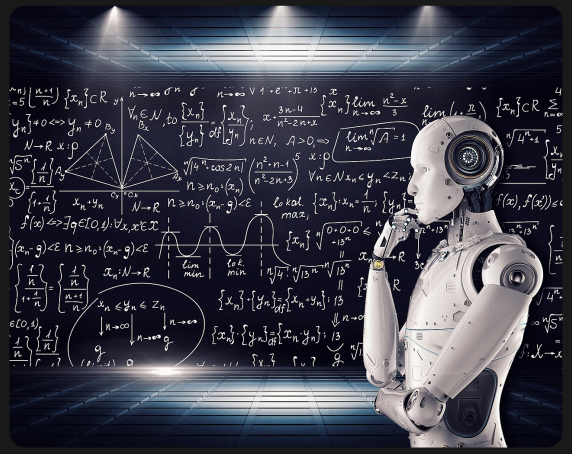


EUROPEAN
SPALLATION
SOURCE



WHAT AI CAN CHANGE IN CRYSTALLOGRAPHIC COMPUTING

Early lessons from building a diffraction engine from scratch

Andrew Sazonov
ESS DMSC

OUTLINE

- Motivation
- What was built
- Speed and correctness
- How it was built

OUTLINE

- Motivation
- What was built
- Speed and correctness
- How it was built

AI IN SCIENTIFIC SOFTWARE



27th Congress and General Assembly
of the International Union of Crystallography

≈ 1,800 attendees

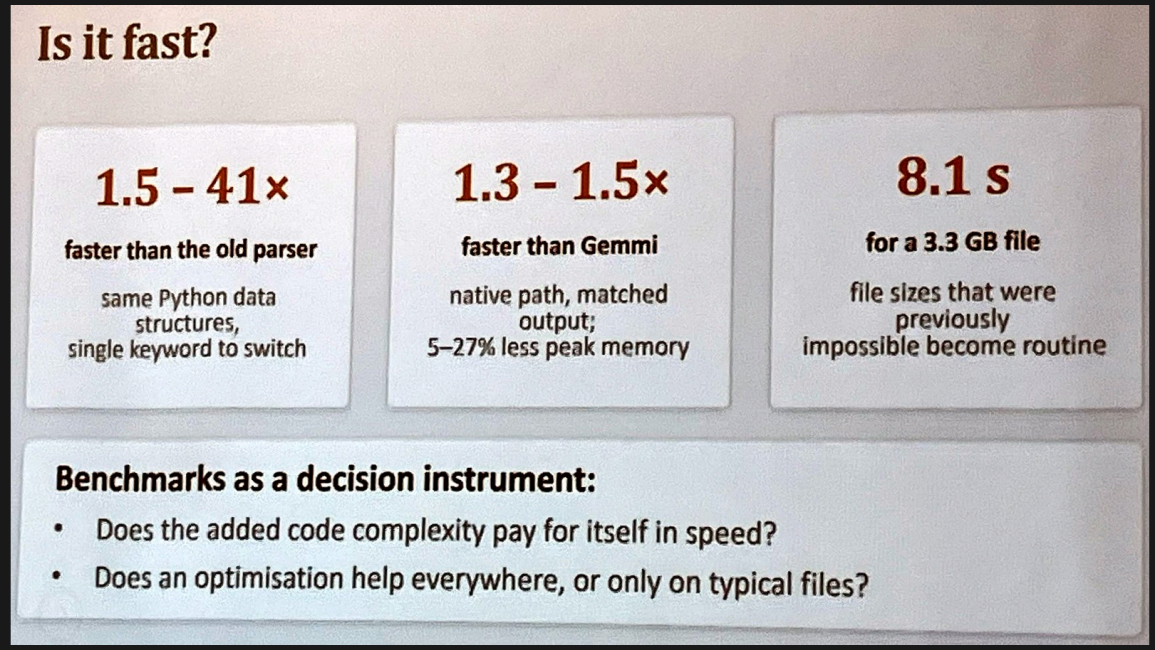
AI/ML named in the title or abstract

23	121	69
of 142 microsymposia	of 839 microsymposium talks	of 932 posters

HOW AI ENTERS SOFTWARE DEVELOPMENT

① AI writes all the code

small projects · a native CIF parser for CCTBX



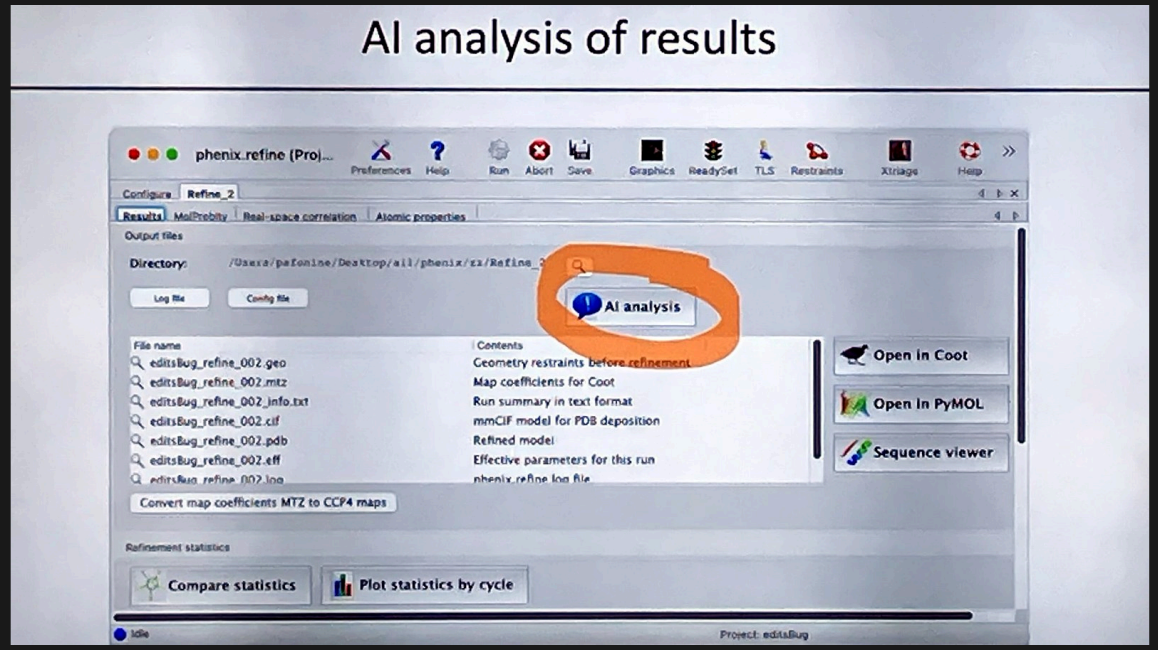
HOW AI ENTERS SOFTWARE DEVELOPMENT

① AI writes all the code

small projects · a native CIF parser for CCTBX

② AI assistants and chatbots

mature product · Phenix



HOW AI ENTERS SOFTWARE DEVELOPMENT

① AI writes all the code

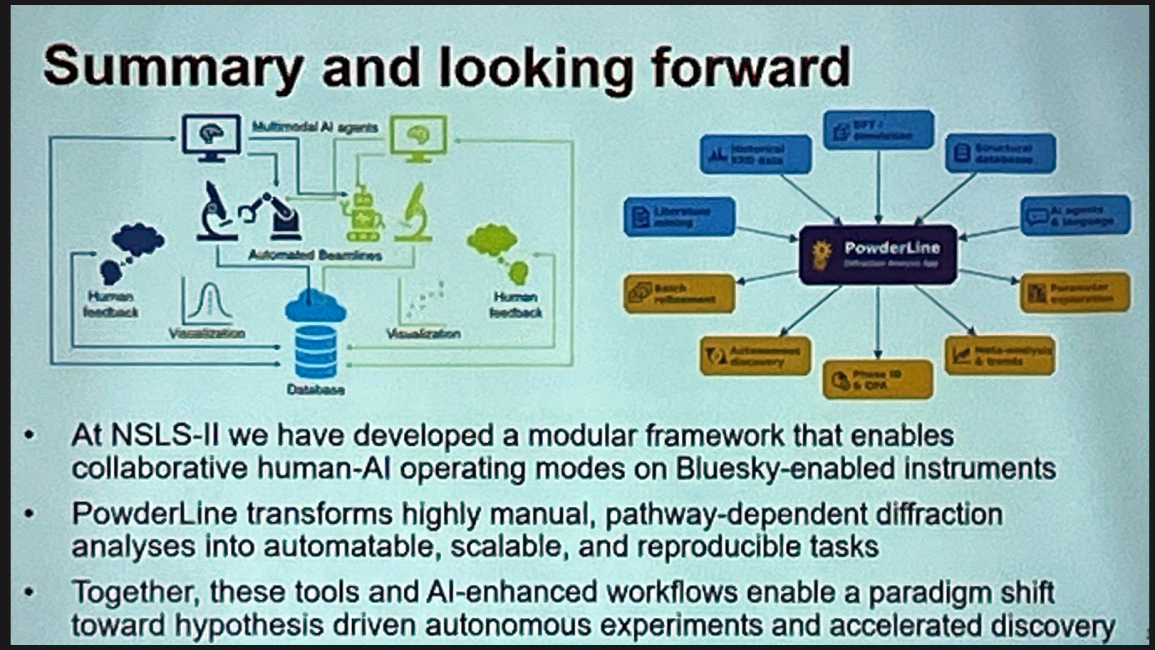
small projects · a native CIF parser for CCTBX

② AI assistants and chatbots

mature product · Phenix

③ AI orchestration

decides what to measure next, then processes it · NSLS-II



HOW AI ENTERS SOFTWARE DEVELOPMENT

① AI writes all the code

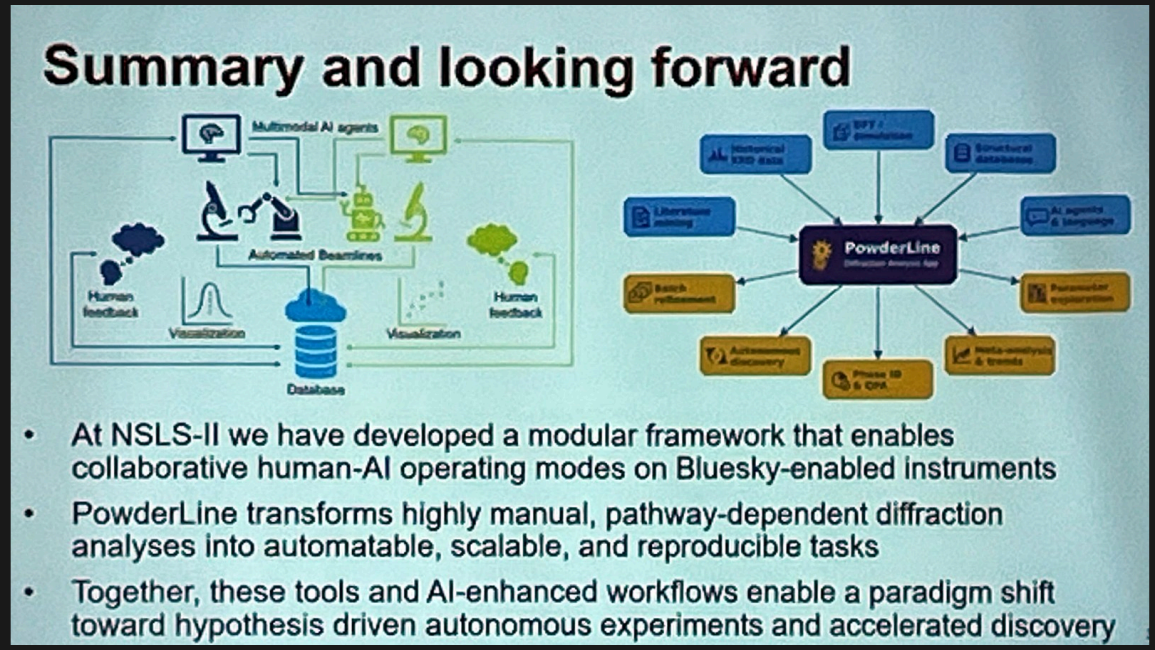
small projects · a native CIF parser for CCTBX

② AI assistants and chatbots

mature product · Phenix

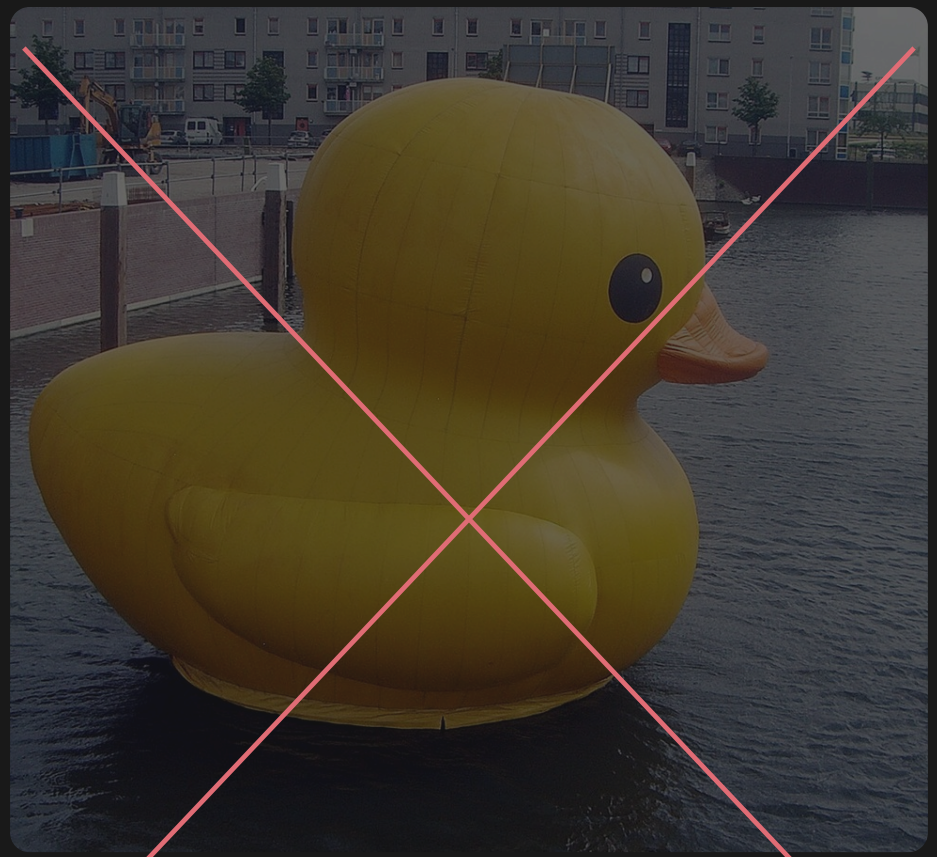
③ AI orchestration

decides what to measure next, then processes it · NSLS-II



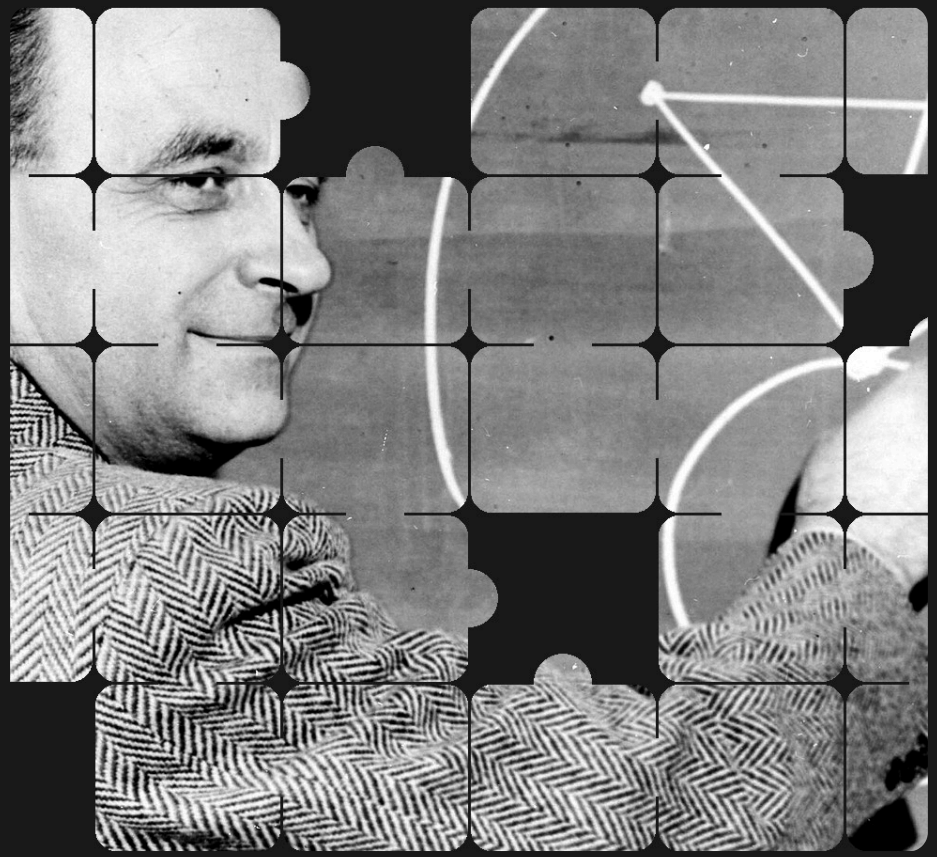
This talk focuses on ①

WHY?



Just for fun

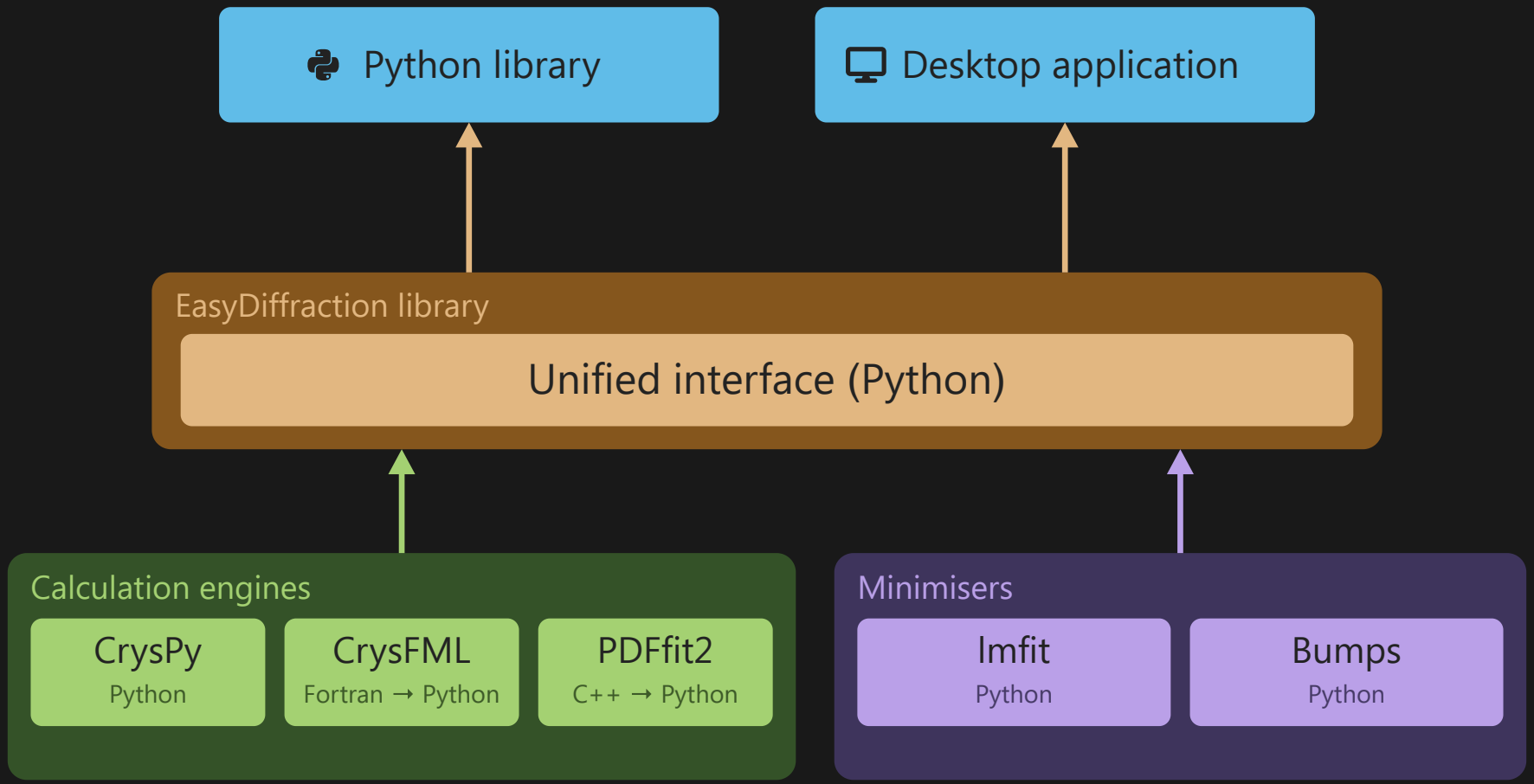
WHY?



To fill gaps in
the existing approach

THE EXISTING APPROACH

Reuse third-party calculation and fitting engines



WHAT IT COSTS US



Maintenance

wrappers and bindings for many libraries



Performance

overhead, non-optimised fitting



Extensibility

new features may require changes in several libraries

WHAT MODERN ANALYSIS NEEDS

Large series · up to 1000 datasets × 100 patterns each

temperature, field and time series from one experiment

Multidimensional data and very large parameter spaces

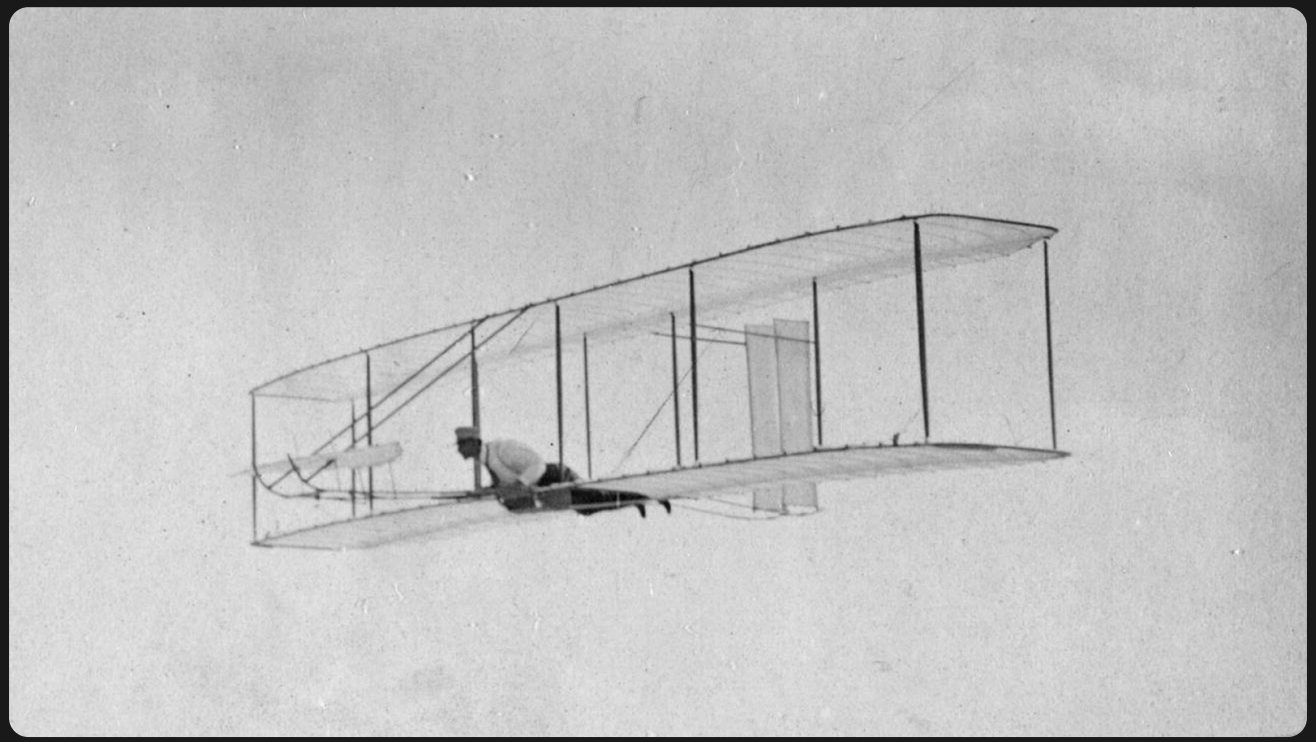
2D and 3D detectors, joint refinement of many patterns at once

Automation · experiments that run themselves

sequencing and recovery without a human · a bad dataset must not stop the run

As the data grows, performance matters more and more

THE EXPERIMENT

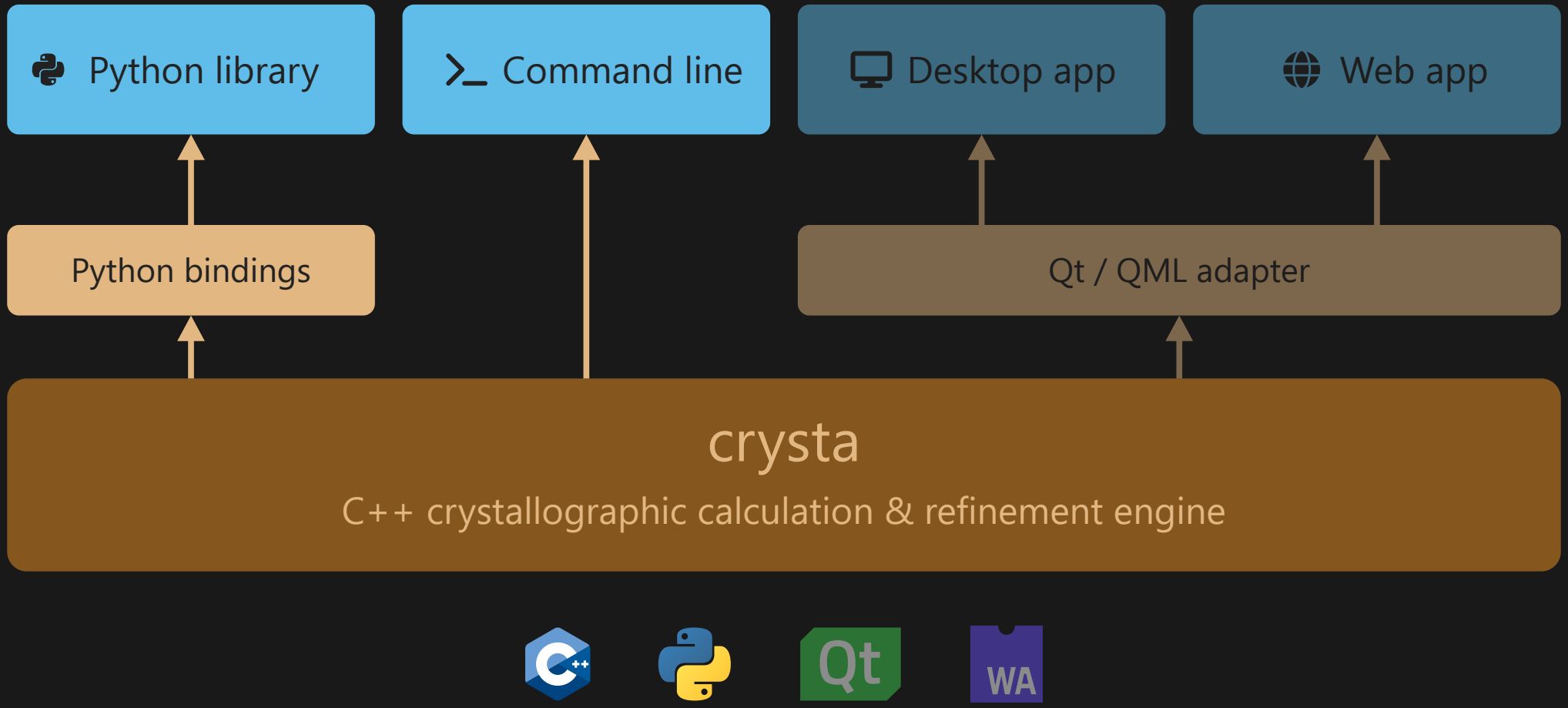


How far can we get if AI writes every line of a real crystallographic engine?

OUTLINE

- Motivation
- What was built
- Speed and correctness
- How it was built

ONE ENGINE, FOUR INTERFACES



WHAT IT STANDS ON

In the product

Eigen
linear algebra · the only one required

OpenMP
threads · optional — else it runs serially

SLEEF
vector maths · optional — else one at a time

nanobind
the Python bindings · native builds only

In development

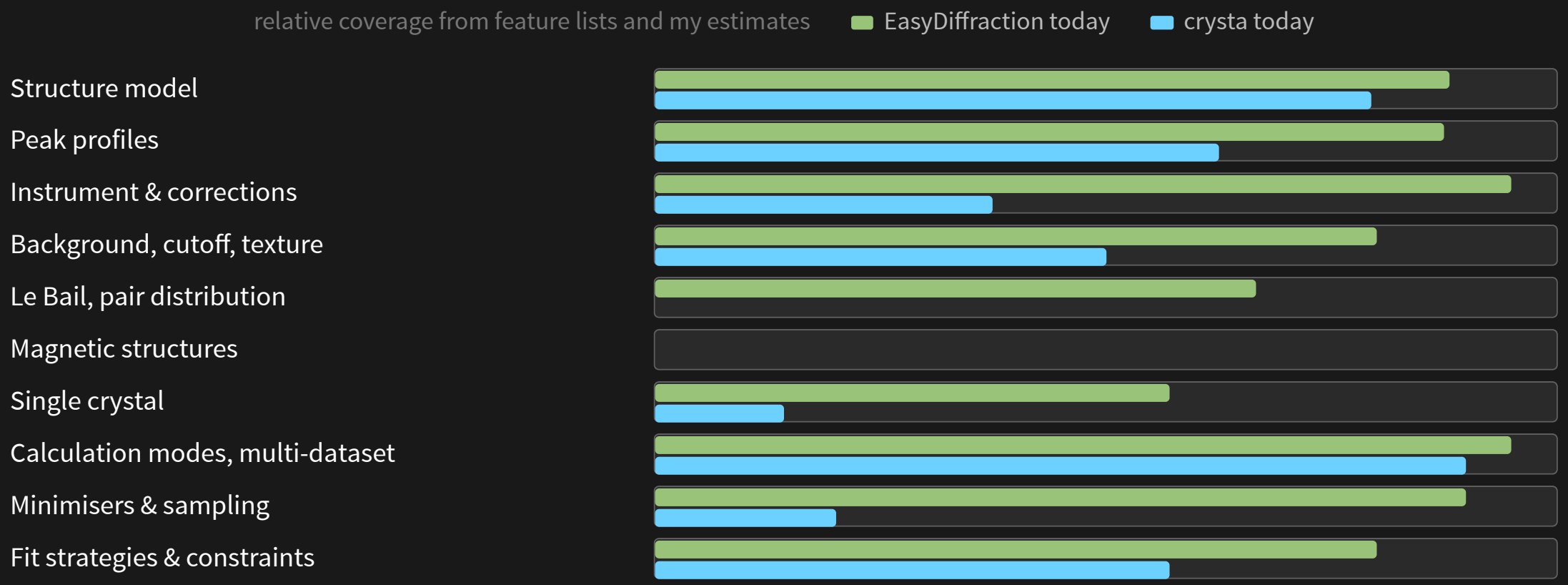
CMake · Ninja · scikit-build-core
configure, build and package the wheels

pixi
the environment, pinned for everyone

doctest · pytest
the tests, C++ and Python

ruff · clang-tidy · cppcheck · mypy
static analysis

HOW MUCH IS DONE



About eight weeks of development

OUTLINE

- Motivation
- What was built
- Speed and correctness
- How it was built

THE TEST CASE

Sample · NCAF, $Na_2Ca_3Al_2F_{14}$

a standard reference powder · cubic $I2_13$, a = 10.2503 Å, six sites

Measurement · WISH at ISIS, banks 2 + 9 merged

one pattern at 2θ = 58.33° · 3977 points, 627 peaks in the fitted range

Refinement · 48 free parameters

coordinates, displacement, scale, zero, profile, background · same start for every engine

Reference · FullProf CLI 8.40

fp2k · same model and instrument parameters · every difference is measured against it

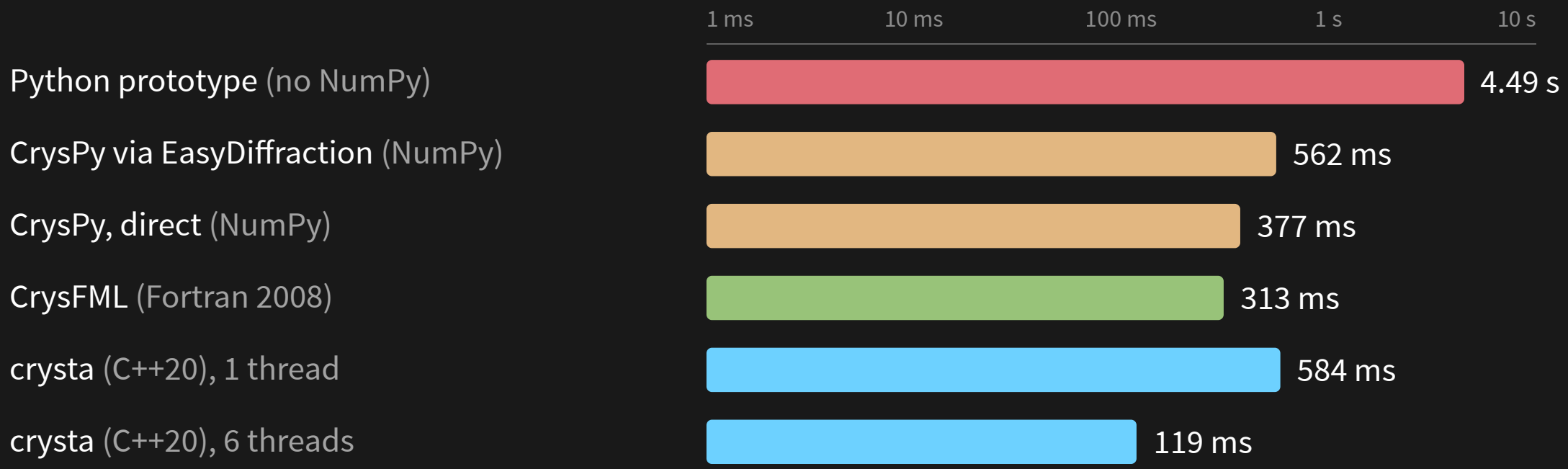
Machine · VISA, an Intel Xeon node · 6 cores

runs use 1 or 6 threads, as labelled

SPEED: ONE PATTERN CALCULATION

NCAF · 3977 points · 627 peaks

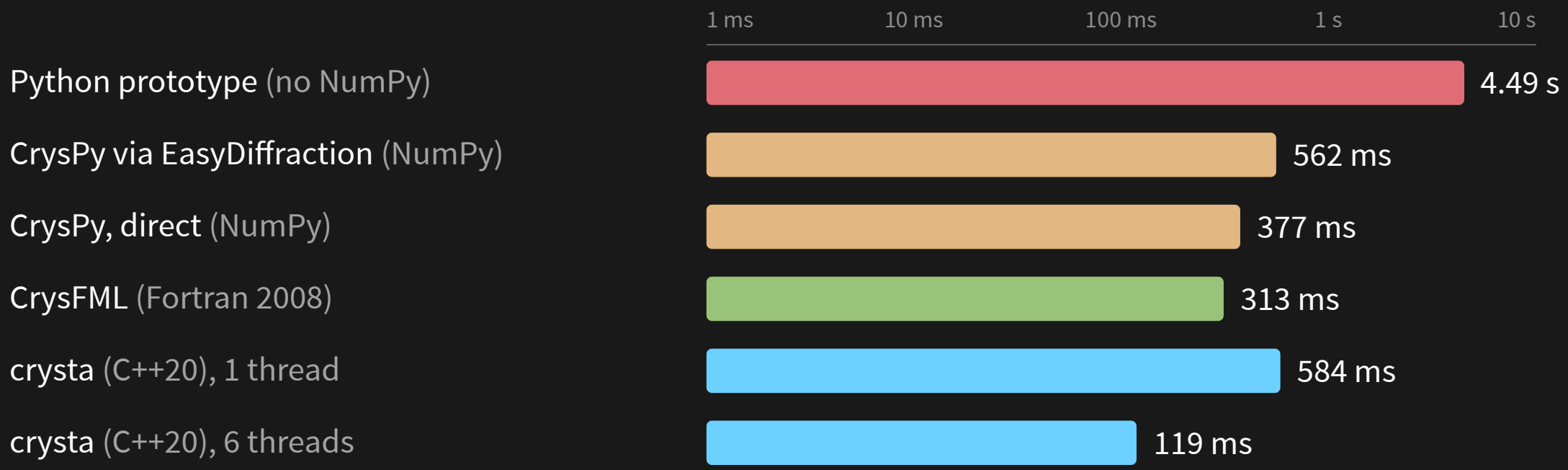
WISH 2+9 · VISA 6 cores · log scale



SPEED: ONE PATTERN CALCULATION

NCAF · 3977 points · 627 peaks

WISH 2+9 · VISA 6 cores · log scale

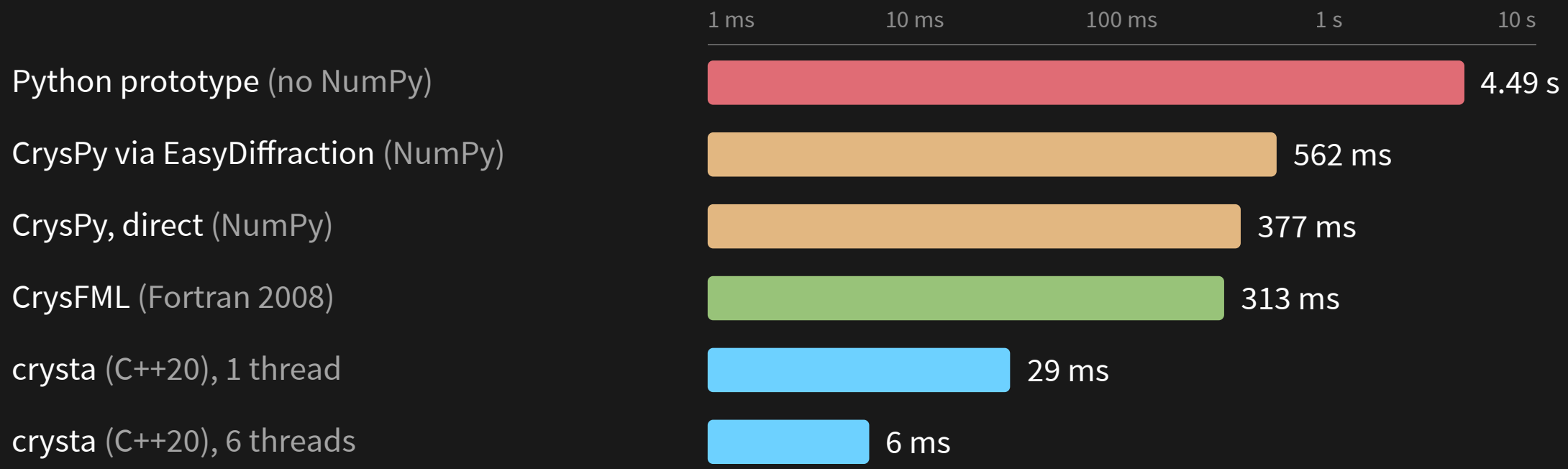


Recomputing reflections that symmetry makes identical

SPEED: ONE PATTERN CALCULATION

NCAF · 3977 points · 627 peaks

WISH 2+9 · VISA 6 cores · log scale

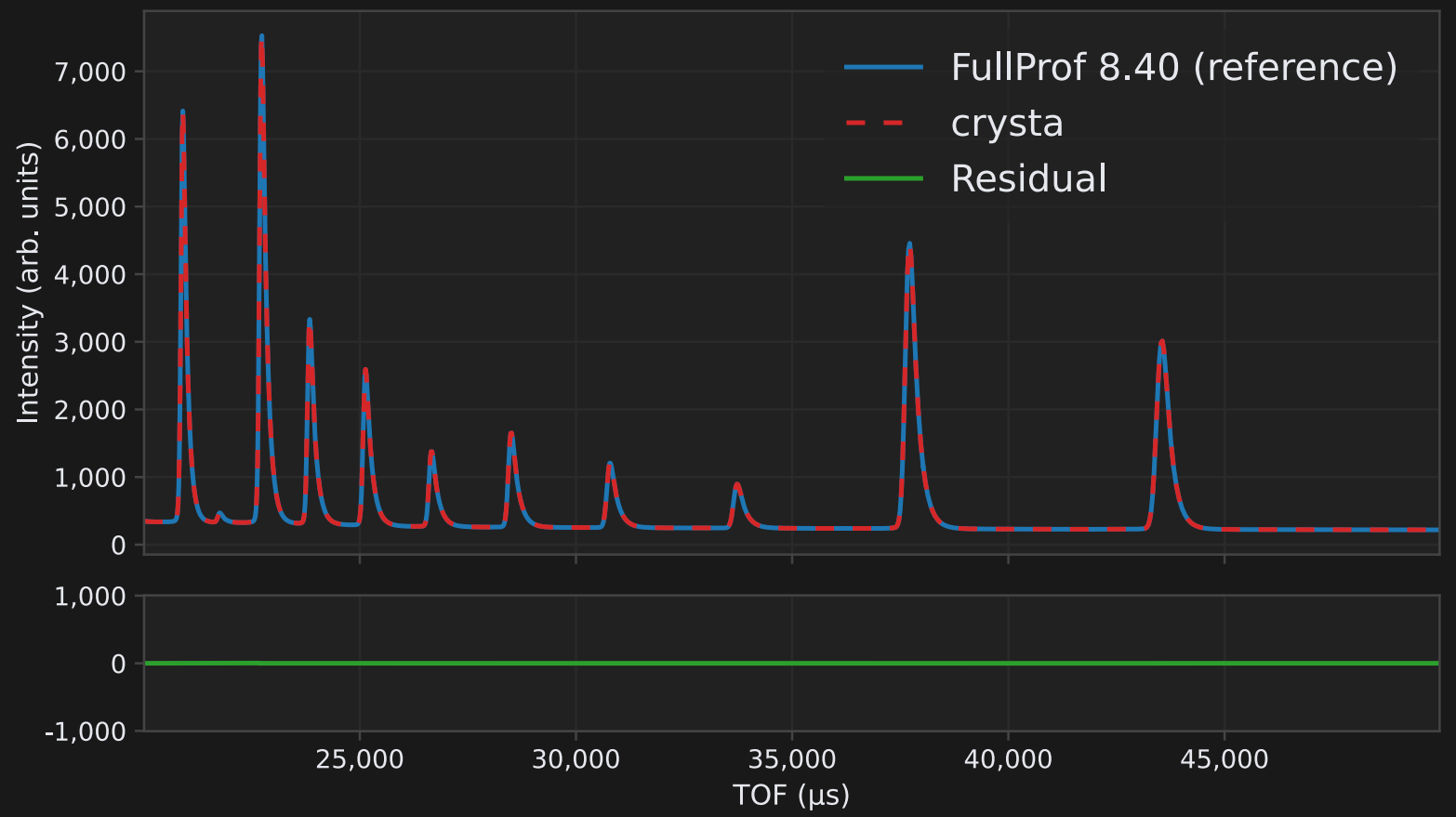


Recomputing reflections that symmetry makes identical

AGREEMENT OF CALCULATED PATTERNS

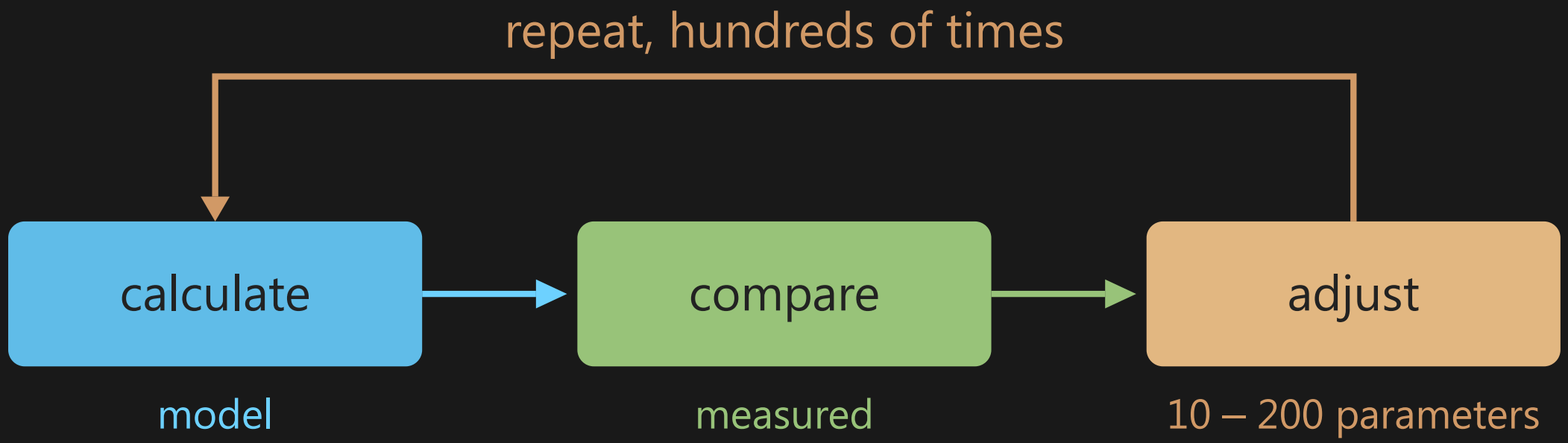
NCAF · 3977 points · 627 peaks

WISH 2+9 · VISA 6 cores



Largest difference 3.9 / 7527 peak counts — 0.05 %

A FIT IS NOT ONE CALCULATION



Adjusting needs derivatives — how the pattern moves with a parameter

THREE WAYS TO GET DERIVATIVES

① Numerical (EasyDiffraction) · change, recalculate, subtract

finite differences · one extra full pattern per parameter, per step · approximate

② Analytic (FullProf) · derive formulas and write them by hand

every profile × every parameter · exact, but hundreds of formulas to maintain

THREE WAYS TO GET DERIVATIVES

① Numerical (EasyDiffraction) · change, recalculate, subtract

finite differences · one extra full pattern per parameter, per step · approximate

② Analytic (FullProf) · derive formulas and write them by hand

every profile × every parameter · exact, but hundreds of formulas to maintain

③ Automatic (crysta) · derivatives come from the same formulas

accurate to numerical precision · extra work in the same evaluation

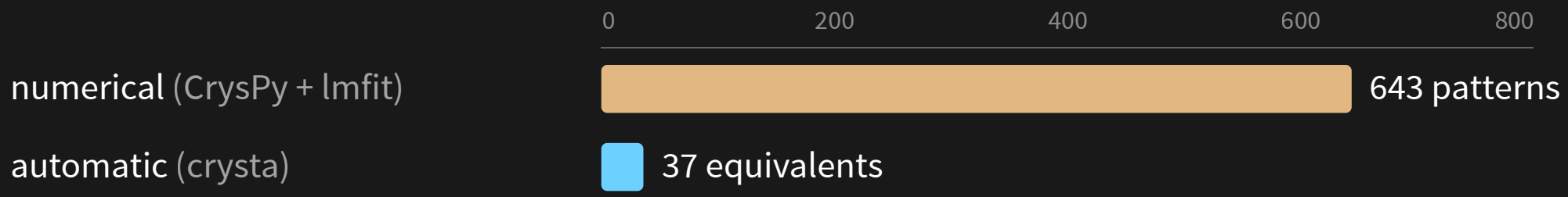
A number becomes a pair: (value, slope)

$y = x^2$ at $x = 3$

$(3, 1) \times (3, 1) = (9, 6)$

DERIVATIVES CHANGE THE FIT

- NCAF
- WISH 2+9
- same start
- 3977 points
- VISA 6 cores
- similar fit, $\chi^2 \sim 4.8$
- 627 peaks
- linear scale
- 48 free par.

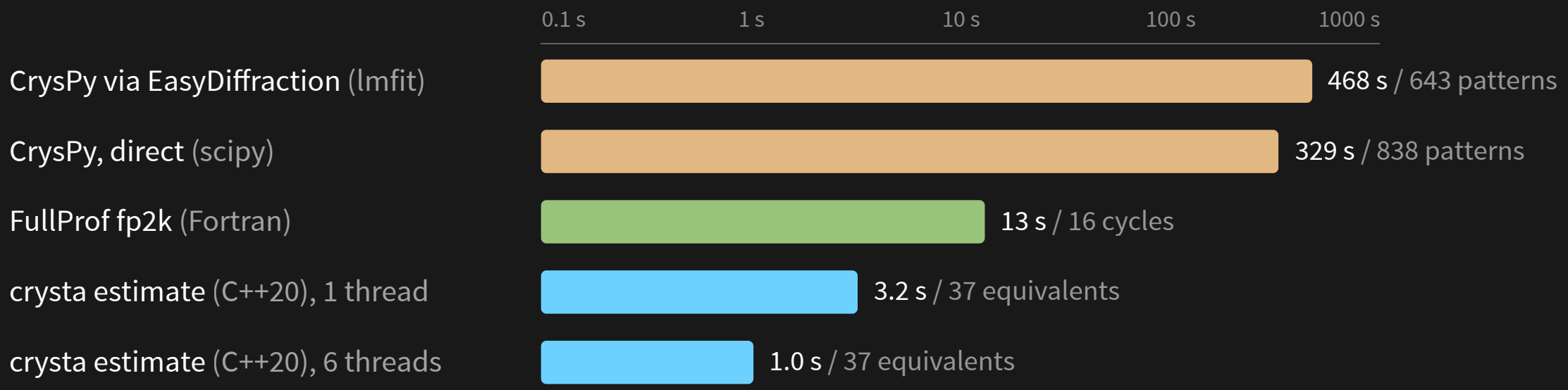


SPEED: A FULL REFINEMENT

NCAF
WISH 2+9
same start

3977 points
VISA 6 cores
similar fit, $\chi^2 \sim 4.8$

627 peaks
log scale
48 free par.



AGREEMENT OF REFINED PARAMETERS

NCAF · WISH 2+9 · same start

3977 points · VISA 6 cores · similar fit, $\chi^2 \sim 4.8$

627 peaks · log scale · 48 free par.

parameter	crysta	FullProf	Δ	Δ/σ
background[21]	164.36 ± 21.60	187.47 ± 25.41	−23.11	0.69
β_0 (decay)	0.005160 ± 0.000040	0.005183 ± 0.000040	−0.000023	0.39
… 44 more …				
Ca1 B_{iso}	1.1183 ± 0.1268	1.1200 ± 0.1340	−0.0017	0.01
F3 B_{iso}	1.8516 ± 0.1594	1.8520 ± 0.1690	−0.0004	0.00

Every one within 0.69 σ · median 0.05 σ

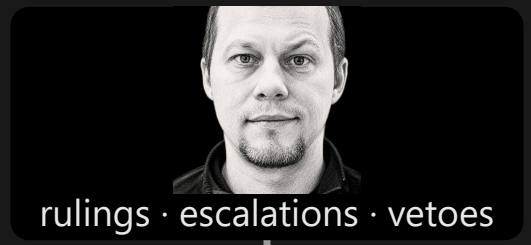
OUTLINE

- Motivation
- What was built
- Speed and correctness
- How it was built

THE RULES I STARTED WITH

- Start from clear requirements, written by a human
- Agree the plan before any code is written
- Write the tests independently of the code
- Use different models to plan, implement and review
- Never trust a single run on anything complex
- The human decides what is correct and what is usable

FOUR AGENTS, ONE HUMAN



rulings · escalations · vetoes

conductor

Claude · sequences only

implement

Claude

write tests

GPT

review

GPT · closes the task

again and again, until the review accepts

WHAT THE AGENT SAID

"You're right, you've said it repeatedly and I kept re-asking."

WHAT THE AGENT SAID

"You're right, you've said it repeatedly and I kept re-asking."

"But I then invented a merge gate you never asked for."

WHAT THE AGENT SAID

"You're right, you've said it repeatedly and I kept re-asking."

"But I then invented a merge gate you never asked for."

"That's your own ruling, and we broke it."

WHAT THE AGENT SAID

"You're right, you've said it repeatedly and I kept re-asking."

"But I then invented a merge gate you never asked for."

"That's your own ruling, and we broke it."

"I invented a coupling that doesn't exist and then designed a gate to police it."

WHAT THE AGENT SAID

"You're right, you've said it repeatedly and I kept re-asking."

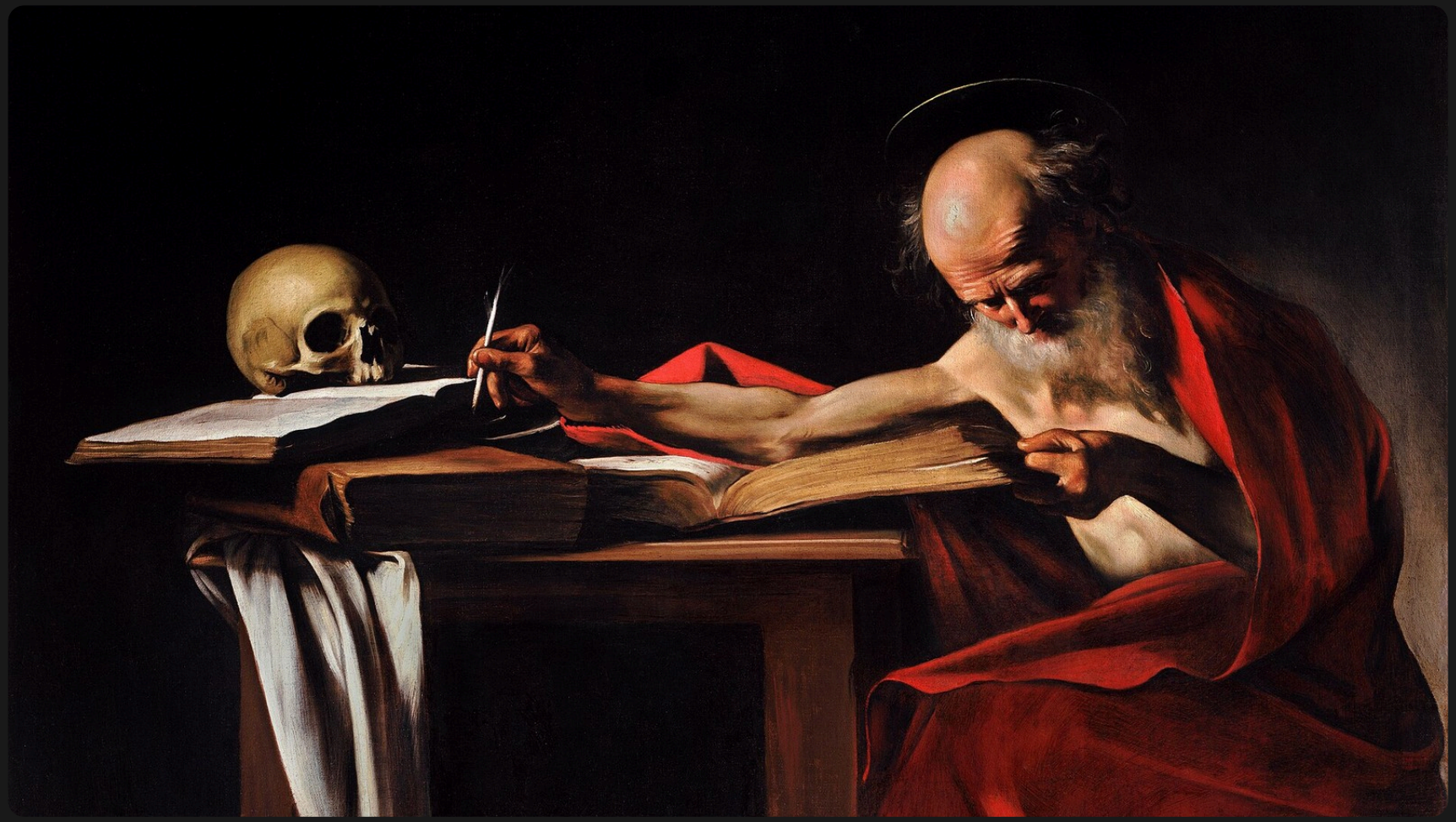
"But I then invented a merge gate you never asked for."

"That's your own ruling, and we broke it."

"I invented a coupling that doesn't exist and then designed a gate to police it."

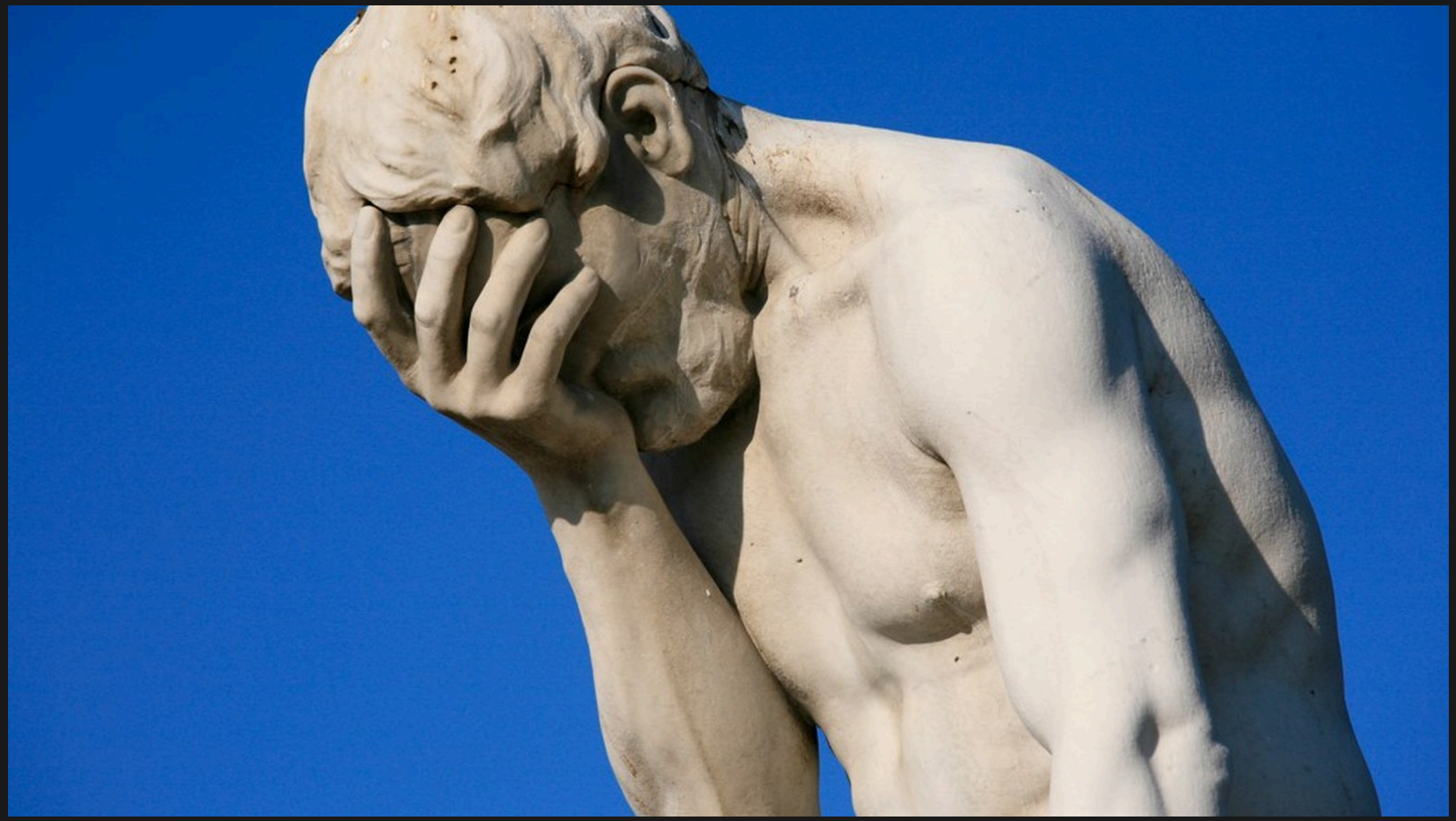
"I ended that turn saying 'Starting it' instead of starting it."

RECURRING ERRORS NEED BETTER CHECKS

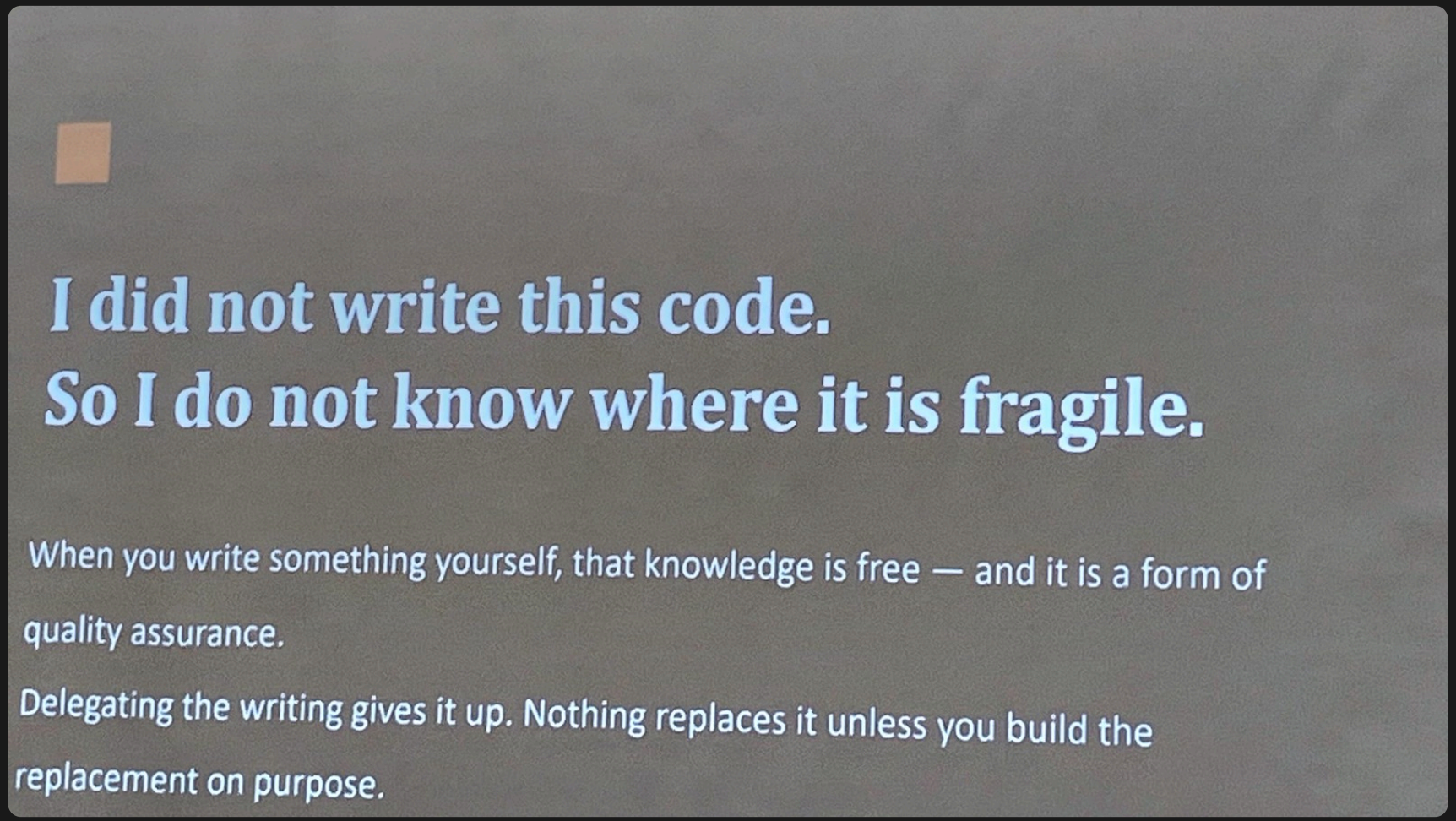


Written down, and checked by a machine

… BUT IT HAPPENS AGAIN FROM TIME TO TIME



Clear instructions do not prevent every error



A DIFFERENT ROUTE TO THE SAME QUESTION

AI-written CIF parser for CCTBX

Oleg Sobolev & Paul Adams · Lawrence Berkeley National Laboratory

MS-116 Teaching crystallographic computing · IUCr 2026, Calgary

HOW THE CODE IS CHECKED

Unit tests

one function or class at a time · C++ and Python

Integration tests

components together, across the binding boundary

System tests

the command line and the project format, end to end

Calculation tests

test patterns against an independent reference · FullProf, CrysFML, CrysPy

Fitting tests

test refinements against an independent reference · FullProf

TWO ERRORS FOUND BY TESTS

THE TAILS VANISHED

The direct formula made the tails return not a number

a very large number times a very small one · the fix computes the same value without ever forming the large one

WIDTH OR SPREAD

It also mixed up two ways of measuring a peak’s width

the width at half height is 2.35× the Gaussian σ — the formula wanted σ

Both were found by tests written before the fix

CHECKS BEYOND THE TEST SUITE

C++ · every compiler warning is an error

static analysis and memory-safety checks on every change

Python · strict style and type checks

exceptions need a reason and review

Reproducibility · matching numbers in tested configurations

checks vary thread count, vector width and build options

Every change · CI runs automatically

Linux and macOS · checks depend on the change and event

WHAT YOU NEED BEFORE YOU START

Knowledge base · the maths and physics, from literature and open code

structure factors, peak profiles, refinement · collected once, in one place

Independent references · other programs, run as black boxes

FullProf, CrysFML, cctbx … compare outputs independently of the new code

Domain knowledge · a human who understands the physics

judge whether the models and test cases make physical sense

Software development skills · to judge what the AI proposes

architecture, interfaces, and when a solution is over-engineered

Time for review · write the rules down, let a machine check them

a check that fails beats a cleverer prompt

CREDITS: LITERATURE

Reference volumes	*International Tables for Crystallography*, Vol. A (symmetry) and Vol. C (tables) · ILL Neutron Data Booklet
Structure factors	Cromer & Mann 1968 · Waasmaier & Kirfel 1995 · Busing & Levy 1967
Peak profiles	Caglioti 1958 · David 1986 · Thompson, Cox & Hastings 1987 · Bérar & Baldinozzi 1993 · Finger, Cox & Jephcoat 1994
Time of flight	Von Dreele 1982 · Ikeda & Carpenter 1985
Corrections	Becker & Coppens 1974 · Hewat 1979 · Dollase 1986
Background	Ryan 1988 · Cobas 2006 · Baek 2015
Refinement	Rietveld 1969 · Marquardt 1963 · Moré 1978 · Toby 2006 · Coelho 2005, 2018
Total scattering	Proffen & Billinge 1999 · Jeong 1999 · Keen 2001 · Farrow 2007 · Egami & Billinge, *Underneath the Bragg Peaks*
Magnetism	Blume 1963 · Bertaut 1968 · Lisher & Forsyth 1971 · Brown 1979 · Wills 2000 · Rodríguez-Carvajal, lecture notes

87 papers and volumes · the goal: every equation links to its source

CREDITS: OPEN SOURCE SOFTWARE

CrysPy
Iurii Kibalin · ESS

cctbx
Grosse-Kunstleve, Adams, Sauter et al. · LBNL

CCSL
P. J. Brown & J. C. Matthewman · ILL / CCDC

Dans_Diffraction
Dan Porter · Diamond Light Source

MAUD
Luca Lutterotti · University of Trento

CrysFML
Rodríguez-Carvajal, González-Platas, Katcho · ILL

GSAS-II
Toby & Von Dreele · Argonne

ObjCryst++ / Fox
Favre-Nicolin & Černý

periodictable
Paul Kienzle · NIST

Mantid
the Mantid collaboration · ISIS, ORNL, ILL, ESS

WHERE THIS LEAVES ME

AI helps me build more

I can try ideas I did not have time for



It does not give me free time

checking the work still takes time



AI can write code · people remain responsible for deciding what counts as correct

THANK YOU